16 B When the sliding section of Path I is physically pushed inwards by a distance of 2.5 cm, the path length of Path I is changed by 5.0 cm = 0.050 m.

DI to CI is half a wavelength → $\frac{\lambda}{2} = 0.050 \text{ m} \Rightarrow \lambda = 0.10 \text{ m}$

Hence, frequency of the wave, $f = \frac{v}{\lambda} = \frac{340}{0.10} = 3.4 \text{ kHz}$